

Grad Café SQL Query Analysis

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Summary of Results

Question	Result
Q1 Fall 2026 applicant count	29,576
Q2 Percent international	46.53%
Q3 Average GPA	3.77
Q3 Average GRE Quantitative	165.83
Q3 Average GRE Verbal	160.70
Q3 Average GRE Analytical Writing	4.35
Q4 Average GPA, American applicants, Fall 2026	3.79
Q5 Fall 2025 acceptance percentage	47.92%
Q6 Average GPA, accepted applicants, Fall 2026	3.76
Q7 JHU Masters in Computer Science entries	8
Q8 Fall 2026 CS PhD acceptances, four universities (original fields)	28
Q9 Same, LLM fields; difference from Q8	28; difference 0
UQ1 Fall 2026 acceptance rate: usable GPA reported / not reported	40.71% / 31.73%
UQ2 Fall 2026 acceptance rate: PhD American / PhD International	25.39% / 26.11%
UQ2 Fall 2026 acceptance rate: Masters American / Masters International	68.64% / 67.45%

Data and Conventions

The database holds 30,000 Grad Café entries added between 1 January and 14 September 2026, scraped and cleaned in Module 2 and loaded into the applicants table by `load_data.py`. Every query below is reproduced exactly as `query_data.py` executes it; where the code interpolates a named constant (a validity range or a matching pattern), the SQL shown has the constant's value filled in.

Text matching is case-insensitive throughout: equality tests compare `LOWER(column)` against a lowercase literal, and pattern tests use PostgreSQL's case-insensitive regular expression operator `~*`. Acceptance means a status of Accepted; the status column holds exactly four values (Accepted, Rejected, Wait listed, Interview) because Module 2 split the decision date into its own column. Percentages are computed in SQL with 100.0 as the first operand, which forces numeric rather than integer division, and `NULLIF` guards every denominator against zero. SQL returns unrounded values; `query_data.py` formats them to the required precision.

Averages include only values on the metric's own scale. An audit of every numeric column, run before any analysis and kept in the repository as `gre_check.sql`, found that the field the site labels GRE holds combined scores on the 260 to 340 scale for 1,406 applicants and quantitative scores on the 130 to 170 scale for only 954, along with values such as 4 and 999 that are not scores on any GRE scale. A value off its metric's scale does not provide that metric, so it is excluded from that average only. The table itself is not modified.

Metric	Valid range	Included	Excluded
GPA	0 to 4.0	17,974	134
GRE Quantitative	130 to 170	954	1,438
GRE Verbal	130 to 170	2,002	4
GRE Analytical Writing	0 to 6	1,807	80

Required Questions

Question 1

Question

How many entries in the database are from applicants who applied for Fall 2026?

Result

Fall 2026 applicant count: 29,576

SQL

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026';
```

Explanation

The query counts every row whose term, lowercased, equals fall 2026. Comparing the lowercased column against a lowercase literal makes the match robust to capitalization while still requiring the whole value to match, so a term such as Fall 2026 matches and Spring 2026 does not. COUNT(*) counts rows rather than non-null values, which is correct here because every row is one applicant entry.

Question 2

Question

Among entries that provide a nationality classification, what percentage are international students?

Result

Percent international: 46.53%

International entries: 13,650 of 29,336 classified entries

SQL

```
SELECT
    COUNT(*) FILTER (WHERE LOWER(us_or_international) = 'international') AS international,
    COUNT(*) AS classified,
    100.0 * COUNT(*) FILTER (WHERE LOWER(us_or_international) = 'international')
        / NULLIF(COUNT(*), 0) AS pct_international
FROM applicants
WHERE us_or_international IS NOT NULL
AND TRIM(us_or_international) <> '';
```

Explanation

The WHERE clause restricts the rows to those with a usable classification, excluding NULL and blank values, so the 664 unclassified entries fall out of the denominator as the question requires. COUNT(*) over the remaining rows is the denominator; COUNT(*) FILTER (WHERE ... = 'international') is the numerator, so American entries (and Other, had any been present) count in the denominator only. The data holds no Other values.

Question 3

Question

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result

Average GPA: 3.77 (n=17,974)

Average GRE Quantitative: 165.83 (n=954)

Average GRE Verbal: 160.70 (n=2,002)

Average GRE Analytical Writing: 4.35 (n=1,807)

SQL

```
SELECT
  AVG(gpa)      FILTER (WHERE gpa BETWEEN 0.0 AND 4.0) AS avg_gpa,
  AVG(gre)      FILTER (WHERE gre BETWEEN 130.0 AND 170.0) AS avg_gre_q,
  AVG(gre_v)    FILTER (WHERE gre_v BETWEEN 130.0 AND 170.0) AS avg_gre_v,
  AVG(gre_aw)   FILTER (WHERE gre_aw BETWEEN 0.0 AND 6.0) AS avg_gre_aw,
  COUNT(gpa)    FILTER (WHERE gpa BETWEEN 0.0 AND 4.0) AS n_gpa,
  COUNT(gre)    FILTER (WHERE gre BETWEEN 130.0 AND 170.0) AS n_gre_q,
  COUNT(gre_v)  FILTER (WHERE gre_v BETWEEN 130.0 AND 170.0) AS n_gre_v,
  COUNT(gre_aw) FILTER (WHERE gre_aw BETWEEN 0.0 AND 6.0) AS n_gre_aw,
  COUNT(gpa) - COUNT(gpa) FILTER (WHERE gpa BETWEEN 0.0 AND 4.0) AS x_gpa,
  COUNT(gre) - COUNT(gre) FILTER (WHERE gre BETWEEN 130.0 AND 170.0) AS x_gre_q,
  COUNT(gre_v) - COUNT(gre_v) FILTER (WHERE gre_v BETWEEN 130.0 AND 170.0) AS x_gre_v,
  COUNT(gre_aw) - COUNT(gre_aw) FILTER (WHERE gre_aw BETWEEN 0.0 AND 6.0) AS x_gre_aw
FROM applicants;
```

Explanation

Each average is computed independently with its own FILTER clause, so an applicant contributes to exactly the averages whose metric they reported: an applicant with a GPA but no GRE score counts toward the GPA average only. This works because missing values were loaded as NULL rather than as empty strings or zeros, and AVG() ignores NULL. The FILTER conditions additionally apply the validity ranges described above. The query also returns, for each metric, how many values were included and how many were excluded as off-scale, so the averages can be read with their sample sizes.

The GRE Quantitative average deserves a caution. It rests on 954 applicants, 3.2% of the dataset, because most applicants who report a GRE entered a combined score rather than a quantitative one.

Question 4

Question

What is the average GPA of American applicants who applied for Fall 2026?

Result

Average GPA of American Fall 2026 applicants: 3.79 (n=10,180)

SQL

```
SELECT AVG(gpa), COUNT(gpa)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
      AND LOWER(us_or_international) = 'american'
      AND gpa BETWEEN 0.0 AND 4.0;
```

Explanation

The three WHERE conditions apply the question's three restrictions simultaneously: the Fall 2026 term, the American classification, and a GPA on the valid scale. A NULL GPA fails the BETWEEN test, so the third condition also enforces that the applicant provides a GPA. AVG() is then taken over the remaining rows, and COUNT(gpa) reports how many contributed.

Question 5

Question

What percentage of Fall 2025 entries are acceptances?

Result

Fall 2025 acceptance percentage: 47.92%

Acceptances: 92 of 192 Fall 2025 entries

SQL

```
SELECT
    COUNT(*) FILTER (WHERE LOWER(status) = 'accepted') AS accepted,
    COUNT(*) AS total,
    100.0 * COUNT(*) FILTER (WHERE LOWER(status) = 'accepted')
      / NULLIF(COUNT(*), 0) AS pct_accepted
FROM applicants
WHERE LOWER(term) = 'fall 2025';
```

Explanation

The WHERE clause restricts the rows to Fall 2025, making COUNT(*) the denominator of all Fall 2025 entries whatever their status, including interviews and waitlists, as the question specifies. The FILTER clause counts acceptances within that set for the numerator.

The denominator is small. The scrape collected entries added in 2026, after the Fall 2025 cycle had largely finished, so these 192 entries are late posters rather than a sample of the Fall 2025 cycle. Their acceptance rate sits well above the 37.10% for Fall 2026 entries (10,974 of 29,576), which is consistent with late posting being more common among applicants with good news.

Question 6

Question

What is the average GPA of accepted applicants who applied for Fall 2026?

Result

Average GPA of accepted Fall 2026 applicants: 3.76 (n=7,204)

SQL

```
SELECT AVG(gpa), COUNT(gpa)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
      AND LOWER(status) = 'accepted'
      AND gpa BETWEEN 0.0 AND 4.0;
```

Explanation

The query applies the three restrictions together: Fall 2026, a status of Accepted, and a GPA on the valid scale, which also excludes applicants who did not report one. It then averages GPA over the remaining rows. The result is effectively identical to the 3.77 average for all applicants who report a GPA, which suggests GPA does little to separate outcomes in a self-reporting population that already clusters near 4.0.

Question 7

Question

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result

JHU Masters in Computer Science count: 8

SQL

```
SELECT COUNT(*)
FROM applicants
WHERE program ~* 'johns? hopkins|mjhu\M'
      AND program ~* 'computer science'
      AND LOWER(degree) = 'masters';
```

Explanation

As the question requires, the query uses the original downloaded fields. The program column holds the combined Program, University text exactly as Grad Café presents it, so both the university and the program are matched within it.

The university pattern `johns? hopkins` matches Johns Hopkins and the common misspelling John Hopkins, and `\mjhu\M` matches the abbreviation JHU. The `\m` and `\M` markers are PostgreSQL's word-start and word-end anchors, so `jhu` matches only as a whole word. Computer Science is matched as a phrase, which also covers longer names such as Computer Science and Engineering. The degree test is an exact case-insensitive match on Masters, which the data confirms is the only master's value present.

Question 8

Question

How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown University, the Massachusetts Institute of Technology, Stanford University, or Carnegie Mellon University?

Result

Fall 2026 CS PhD acceptances at the four universities (original fields): 28

SQL

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
  AND LOWER(status) = 'accepted'
  AND LOWER(degree) = 'phd'
  AND program ~* 'computer science'
  AND program ~* 'georgetown|massachusetts institute of technology|\mmit|M|stanford|carnegie mellon|\mcmu\M';
```

Explanation

All five restrictions apply simultaneously in one WHERE clause: term, acceptance status, and PhD degree by exact case-insensitive equality, and Computer Science and the four universities by pattern against the original program text.

The university pattern accepts each institution's full name plus the abbreviations MIT and CMU. The word anchors around the abbreviations matter: without them, a bare mit would match inside unrelated text such as Smith College, and a test with exactly that record confirmed the anchored pattern excludes it.

Question 9

Question

Repeating Question 8, but identifying the university and program from llm_generated_university and llm_generated_program instead of the original fields, how many entries match, and how does the count differ from Question 8?

Result

Original-field count: 28

LLM-field count: 28

Difference: 0

SQL

Question 9 count, from query_data.py:

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026'
  AND LOWER(status) = 'accepted'
  AND LOWER(degree) = 'phd'
  AND llm_generated_program ~* 'computer science'
  AND llm_generated_university ~* 'georgetown|massachusetts institute of technology|\mmit|M|stanford|carnegie mellon|\mcmu\M';
```

Record-level agreement check, from questions_scratch.sql:

```
SELECT (program ~* 'computer science'
        AND program ~* 'georgetown|massachusetts institute of technology|\mmit|M|stanford|carnegie mellon|\mcmu\M') AS original_match,
       (llm_generated_program ~* 'computer science'
        AND llm_generated_university ~* 'georgetown|massachusetts institute of technology|\mmit|M|stanford|carnegie mellon|\mcmu\M') AS llm_match,
       COUNT(*)
FROM applicants
WHERE LOWER(term) = 'fall 2026' AND LOWER(status) = 'accepted' AND LOWER(degree) = 'phd'
GROUP BY 1, 2 ORDER BY 1, 2;
```

Explanation

The first query is identical to Question 8 except that the program and university patterns are applied to the LLM-generated columns. Term, degree, and status stay on the original fields, as the question specifies. query_data.py runs both counts and subtracts them to report the difference.

Equal totals do not by themselves show that the two methods selected the same records: offsetting disagreements, one record gained and another lost, would also produce a difference of 0. The second query checks this directly. Under the shared term, status, and degree filters, it cross-tabulates the original-field match against the LLM-field match. It returned 28 records matched by both methods and none matched by only one, so the methods agree record for record.

The agreement reflects this question more than the reliability of the LLM fields. Module 2's accuracy analysis found that the standardizer's confabulations cluster on short, unqualified institution names, for example completing University of Michigan as University of Milan in 287 records. The four universities here have distinctive full names that leave the model nothing to complete, and Computer Science is equally unambiguous. A question about institutions with generic names would likely have produced differing counts.

Original Questions

User Question 1

Question

Among Fall 2026 entries, do applicants who report a usable GPA post acceptances at a different rate from applicants who do not?

Result

Reported GPA: 40.71% (7,204 of 17,695)

No usable GPA: 31.73% (3,770 of 11,881)

SQL

```
SELECT
  CASE WHEN gpa BETWEEN 0.0 AND 4.0
        THEN 'Reported GPA' ELSE 'No usable GPA' END AS gpa_reported,
  COUNT(*) AS entries,
  COUNT(*) FILTER (WHERE LOWER(status) = 'accepted') AS accepted,
  100.0 * COUNT(*) FILTER (WHERE LOWER(status) = 'accepted')
    / NULLIF(COUNT(*), 0) AS pct_accepted
FROM applicants
WHERE LOWER(term) = 'fall 2026'
GROUP BY 1
ORDER BY 1 DESC;
```

Explanation

The CASE expression sorts every Fall 2026 entry into one of two groups according to whether it has a GPA on the valid scale, and GROUP BY computes an acceptance rate for each group. The rate uses the same definition as Question 5, acceptances over all entries in the group, so the figures are comparable.

The question tests whether self-reporting is correlated with outcome. It is: applicants who share a GPA post acceptance 9% more often than those who do not. Either accepted applicants are more willing to share their credentials, or rejected applicants are less willing, and both readings mean that averages computed over reported values describe a population tilted toward acceptances.

User Question 2

Question

How do Fall 2026 acceptance rates differ between American and International applicants, for PhD and Masters programs?

Result

PhD, American: 25.39% (2,546 of 10,029)

PhD, International: 26.11% (2,781 of 10,650)

Masters, American: 68.64% (3,021 of 4,401)

Masters, International: 67.45% (1,846 of 2,737)

SQL

```
SELECT
  CASE LOWER(degree) WHEN 'phd' THEN 'PhD' ELSE 'Masters' END AS degree_group,
  CASE LOWER(us_or_international) WHEN 'american' THEN 'American'
    ELSE 'International' END AS nationality,
  COUNT(*) AS entries,
  COUNT(*) FILTER (WHERE LOWER(status) = 'accepted') AS accepted,
  100.0 * COUNT(*) FILTER (WHERE LOWER(status) = 'accepted')
    / NULLIF(COUNT(*), 0) AS pct_accepted
FROM applicants
WHERE LOWER(term) = 'fall 2026'
  AND LOWER(degree) IN ('phd', 'masters')
  AND LOWER(us_or_international) IN ('american', 'international')
GROUP BY 1, 2
ORDER BY 1 DESC, 2;
```

Explanation

The query restricts the rows to Fall 2026 PhD and Masters entries with an American or International classification, then groups by degree and nationality and computes the acceptance rate for each of the four combinations, using the same definition as Question 5. The CASE expressions normalize capitalization before grouping, so that variants such as masters and Masters cannot form separate groups.

Degree type dominates: Masters entries report acceptances at roughly 68% against roughly 26% for PhD entries, while nationality moves the rate by little more than a point within either degree. A pooled acceptance rate over this dataset therefore depends largely on its mix of degree types, which is itself 70.7% PhD.